```
adjusted_mutual_info_score
( labels_true, labels, average_method='arithmetic'))
print("Silhouette Coefficient: %0.3f"  % metrics.silhouette_
score(X, labels))

##############################################################
# Plot result
import matplotlib.pyplot as plt

# Black removed and is used for noise instead.
unique_labels = set(labels)
colors = [plt.cm.Spectral(each)
          for each in np.linspace(0, 1, len(unique_labels))]
for k, col in zip(unique_labels, colors):
    if k == -1:
        col = [0, 0, 0, 1]
    class_member_mask = (labels == k)
    xy = X[class_member_mask & core_samples_mask]
    plt.plot(xy[:, 0], xy[:, 1], 'o',
markerfacecolor=tuple(col), markeredgecolor='k',
markersize=14)
    xy = X[class_member_mask & ~core_samples_mask]
    plt.plot(xy[:, 0], xy[:, 1], 'o', markerfacecolor=tuple(c
ol),markeredgecolor='k',  markersize=6)

plt.title('Estimated number of clusters: %d'  % n_clusters)
plt.show()
```

The output is:

```
Estimated number of clusters:  3
Estimated number of noise points:  18
Homogeneity: 0.953
Completeness: 0.883
V-measure: 0.917
Adjusted Rand Index: 0.952
Adjusted Mutual Information: 0.916
Silhouette Coefficient: 0.626
```

To wrap up, we went through some basic concepts of the k-means and DBSCAN algorithms. A direct comparison with k-means clustering has been made to better understand

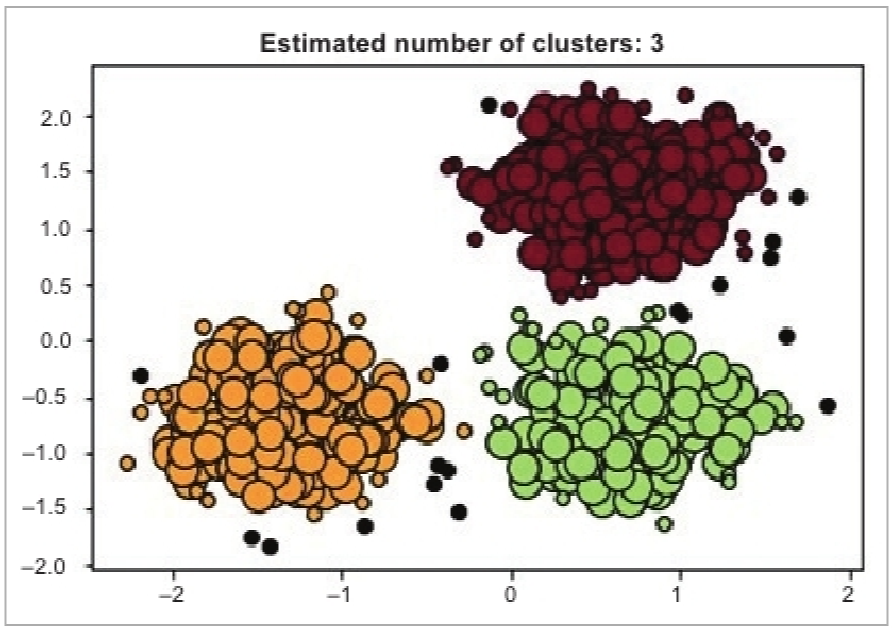

Figure 7: Results of the DBSCAN clustering algorithm

the differences between these algorithms. Basically, the DBSCAN algorithm overcomes the drawbacks of the k-means algorithm.

Clustering is an unsupervised machine learning approach, but can be used to improve the accuracy of supervised machine learning algorithms. Hopefully, this will help you to get started with one of the most used clustering algorithms for unsupervised problems. END

References

[1] *https://towardsdatascience.com/the-5-clustering-algorithms-data-scientists-need-to-know-a36d136ef68*
[2] *https://www.geeksforgeeks.org/dbscan-clustering-in-ml-density-based-clustering/*
[3] *https://www.coursera.org/learn/machine-learning-with-python/*
[4] *https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html*

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